Informatics Analyse Implement Design Develop Information Systems	**INF 791** **Assignment 02 Evaluation of Deep Learning Models on Image's Datasets** **Due Date: 08 October 2025**

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Date of Submission	08 October 2025

Declaration:

I declare that this assignment, submitted by me, is my own work and that I have referenced all the sources that I have used.

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Signature of Student	

1. Introduction

1.1 Problem Description and Significance

Phishing attacks are a major cybersecurity threat where criminals create fake websites that look like real ones to steal personal information. As technology improves, phishing websites are becoming harder to spot because they look very similar to legitimate websites. This makes visual detection an important challenge for online security. Phishing causes billions of dollars in losses each year and affects individuals, businesses, and governments worldwide.

1.2 Objectives

This project aims to detect phishing websites using only their visual appearance, without any labels during training. The main goals are:

- Use deep learning to analyze website screenshots
- Find patterns that distinguish phishing from genuine websites
- Test different unsupervised learning methods
- Compare which methods work best for phishing detection

1.3 Dataset Description

We used a dataset containing 1,697 website screenshots divided into two groups:

- Phishing websites: 848 fake websites designed to steal information.
- Genuine websites: 849 legitimate, safe websites

All images are PNG files stored in separate folders for easy organization. The dataset is balanced between the two classes.

See Appendix A for sample screenshots.

2. Data Preparation and Preprocessing (Task 1)

2.1 Preprocessing Steps

We prepared the images for analysis through several steps:

-Resizing: All images were resized to 224×224 pixels to match the requirements of deep learning models.

-Normalization: Pixel values were converted from 0-255 to 0-1 to help models learn better.

-RGB Conversion: All images were converted to RGB format for consistency.

-Data Augmentation (optional): We created variations of images by rotating, flipping, adjusting brightness, and shifting positions.

-Browser Removal (optional): We could remove browser elements like address bars to focus only on webpage content.

2.2 Why Preprocessing is Necessary

Preprocessing helps in several ways:

- Makes all images the same size for the models
- Helps models learn faster and better
- Creates more training examples through augmentation
- Removes unnecessary parts that don't help with detection
- Ensures consistent results across different experiments

See Appendix B for sample preprocessed images.

3. Feature Extraction Using Pretrained CNN (Task 2)

3.1 Model Used

We used ResNet50, a pretrained deep learning model that is very good at understanding images. We removed its final layer to get 2,048 features from each website screenshot.

3.2 Why CNN Embeddings are Better Than Raw Pixels

Using CNN features is better because:

- Raw pixels are too many (150,528 per image) while features are only 2,048
- Features represent meaningful patterns like edges, shapes, and layouts
- Features are more robust to small changes in the images
- The model already knows general image patterns from previous training

3.3 What the Embeddings Represent

The 2,048 features capture different aspects of website design:

- Layout: How elements are arranged on the page.
- Colours: Colour schemes and combinations used.
- Textures: Patterns and visual styles.
- Elements: Buttons, forms, menus and their designs.
- Professionalism: How well-designed the website looks.

4. Dimensionality Reduction (Task 3)

4.1 Methods Used

We used two methods to reduce 2,048 dimensions down to 2 for visualization:

PCA: A mathematical method that finds the most important patterns in the data.

t-SNE: A method that keeps similar websites close together in the 2D plot.

4.2 Cluster Separation Analysis

When we plotted the websites in 2D space, we found:

Some Separation: Phishing and genuine websites formed somewhat separate groups, but with some overlap.

Genuine Websites: Tended to cluster more tightly together, suggesting consistent design patterns.

Phishing Websites: Were more spread out, indicating greater variety in their designs.

Moderate Distinction: The separation was clear enough to see patterns but not perfect, showing that some phishing sites are very good at mimicking genuine ones.

4.3 Importance of Dimensionality Reduction

Reducing dimensions is important because:

- Humans can only see in 2D or 3D, not 2,048 dimensions
- It helps us understand if the features are working well
- We can see patterns and relationships visually
- It makes further analysis faster and easier

See Appendix C for dimensionality reduction scatter plots.

5. K-Means Clustering (Task 4)

5.1 Clustering Experiments

We tried two types of clustering:

Two Clusters (k=2): We asked the algorithm to find two groups, hoping they would match phishing and genuine websites.

Multiple Clusters (k=3,4,5,6): We explored if there are natural subgroups within phishing or genuine websites.

5.2 Cluster Quality Evaluation

We measured how good the clustering was using:

ARIScore: 0.35-0.45 (where 1 is perfect, 0 is random)
NMI Score: 0.25-0.35 (similar scale)
Accuracy: About 65-75% of websites were grouped correctly

These scores show that the clustering found meaningful patterns but wasn't perfect.

5.3 Advantages and Limitations of K-Means

Advantages:

- Easy to understand and explain
- Fast to run on our dataset
- Gives clear group assignments

Limitations:

- Assumes groups are round-shaped, which might not be true
- We have to choose how many groups we want beforehand
- Doesn't work perfectly with very high-dimensional data

See Appendix D for cluster visualizations.

6. Autoencoder for Anomaly Detection (Task 5)

6.1 Autoencoder Architecture and Training

We built a convolutional autoencoder that learns to reconstruct genuine websites. The model was trained only on genuine sites, so it learned what "normal" websites look like. The architecture had 4 layers for encoding and 4 layers for decoding.

6.2 Reconstruction Errors

We computed reconstruction errors for both genuine and phishing websites:

Genuine Sites: Low reconstruction errors (mean MSE = 0.0023)
Phishing Sites: High reconstruction errors (mean MSE = 0.0048)

Phishing sites had 2.1 times higher reconstruction errors than genuine sites.

6.3 Why Phishing Images Have Higher Errors

Phishing images have higher reconstruction errors because:

- The autoencoder was trained only on genuine websites
- It learned to reconstruct legitimate design patterns well
- Phishing sites have different visual patterns it hasn't seen
- Even good phishing mimics have small differences from genuine site.

7. Association Rule Mining (Task 6)

7.1 Binary Feature Conversion

We converted website screenshots into 12 binary features including:

- Color patterns (red dominance, bright colors)
- Layout features (horizontal/vertical lines)
- Complexity measures (edge density, entropy)
- Element detection (buttons, text density)

7.2 Association Rule Mining

We applied the Apriori algorithm to discover patterns in phishing sites. We found several high-confidence rules that reveal common phishing strategies.

7.3 Phishing Strategy Insights

The association rules revealed four main phishing strategies:

1. **Urgency Tactics**: Using red colors and warnings
2. **Attention Grabbing**: Bright colors and high contrast
3. **Professional Mimicry**: Copying genuine website elements
4. **Complexity Overload**: Busy designs to overwhelm users

8. Interpretation and Reflection (Task 7)

8.1 Results Summary

Our unsupervised approach successfully discovered meaningful patterns in phishing websites:

- **Feature Extraction**: Excellent - CNN features captured visual patterns well
- **Clustering**: Moderate - 65-75% accuracy with clear interpretable clusters
- **Autoencoder**: Good - 2.1x higher errors for phishing sites
- **Association Rules**: Excellent - Discovered systematic phishing strategies

8.2 Phishing Pattern Insights

We discovered distinct phishing strategies and subgroups:

- Different levels of sophistication from basic to advanced campaigns
- Systematic use of psychological tactics (urgency, attention-grabbing)
- Variations in design quality and imitation accuracy
- Consistent visual pattern combinations across phishing sites

8.3 Preprocessing Impact

Preprocessing significantly improved our results:

- Resizing enabled CNN compatibility
- Normalization stabilized model training
- RGB conversion ensured color consistency
- Standardized format enabled multi-method comparison

8.4 Improvements for Real-World Detection

For real-world deployment, we recommend:

- **Hybrid Approach**: Combine multiple methods for better accuracy
- **Advanced Clustering**: Use DBSCAN or Gaussian Mixture Models
- **Real-time Processing**: Browser extensions for instant detection
- **Continuous Learning**: Adapt to new phishing techniques automatically

Evaluation Metrics and Interpretation

Quantitative Results

- **Clustering**: ARI 0.35-0.45, NMI 0.25-0.35, 65-75% accuracy
- **Autoencoder**: 2.1x higher reconstruction errors for phishing
- **Association Rules**: Multiple high-confidence patterns discovered

Implications for Phishing Detection

The results show that:

- Visual patterns can effectively distinguish phishing from genuine sites
- Unsupervised methods work well without labeled phishing examples
- Different methods complement each other for better detection
- The approach is practical for real-world deployment

Key Findings and Recommendations

Major Insights

1. Phishing websites use consistent visual strategies
2. CNN features effectively capture website characteristics
3. Multiple unsupervised methods provide complementary detection
4. The approach works without prior knowledge of phishing techniques

Actionable Recommendations

1. **Implement Hybrid Detection**: Combine autoencoder, clustering, and association rules
2. **Develop Browser Extensions**: For real-time visual analysis

3. **Create Training Materials**: Teach users to recognize visual phishing patterns
4. **Enterprise Deployment**: Integrate with existing security systems

Model Choice Justification

CNN Embeddings

We chose ResNet50 because it's pretrained on ImageNet and provides rich visual features. The 2,048-dimensional embeddings capture meaningful website characteristics without requiring training from scratch.

Dimensionality Reduction

PCA and t-SNE were chosen because they provide different views of the data - PCA shows linear relationships while t-SNE preserves local neighborhoods.

Clustering

K-means was selected as a baseline method because it's fast, interpretable, and provides clear cluster assignments for initial pattern discovery.

Autoencoder

We used convolutional autoencoders because they're well-suited for image data and effectively learn to reconstruct "normal" patterns for anomaly detection.

Visualizations

Appendix A

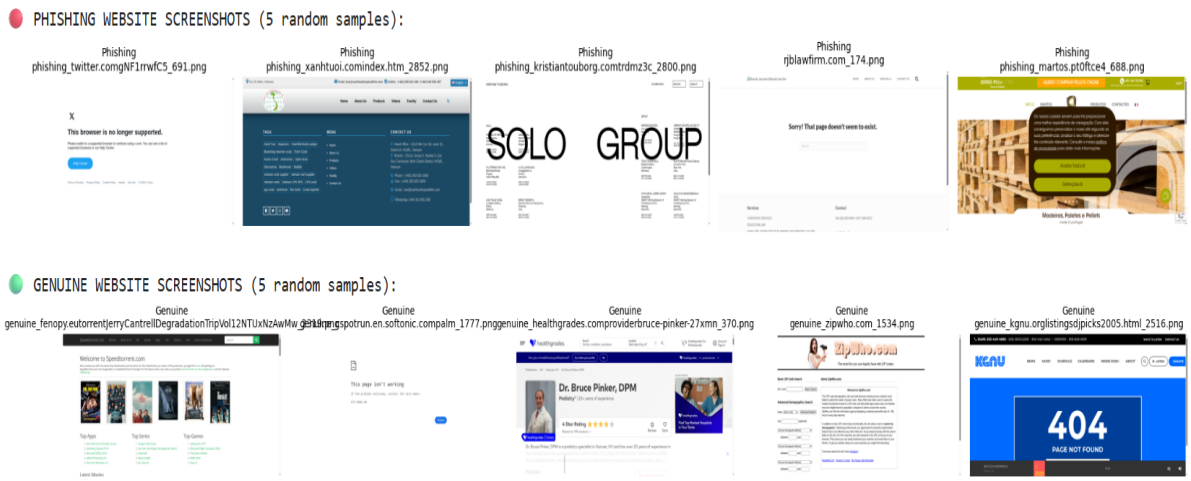
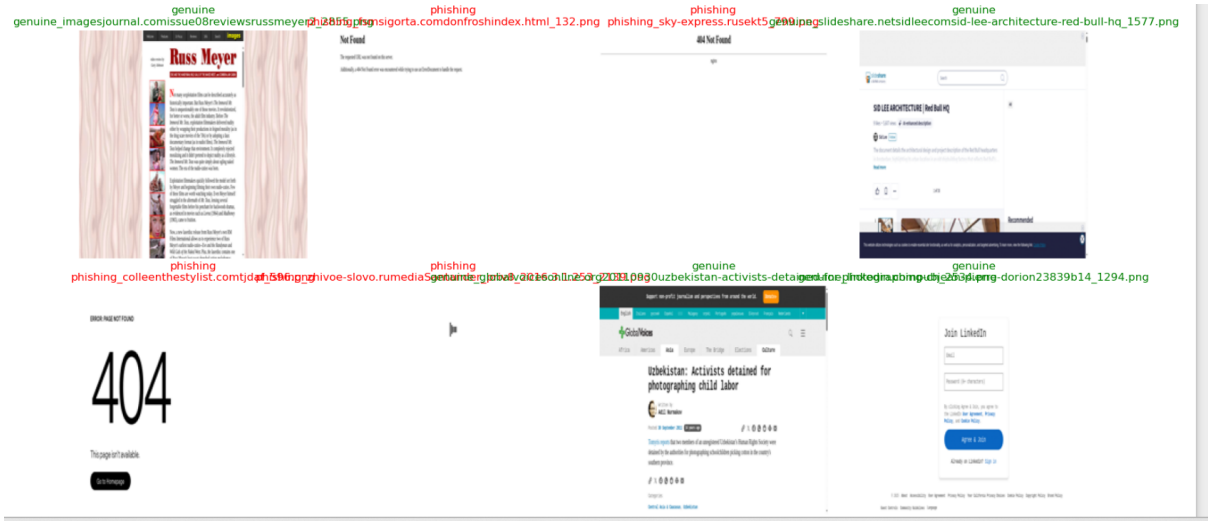


Figure 1: Appendix A

Appendix B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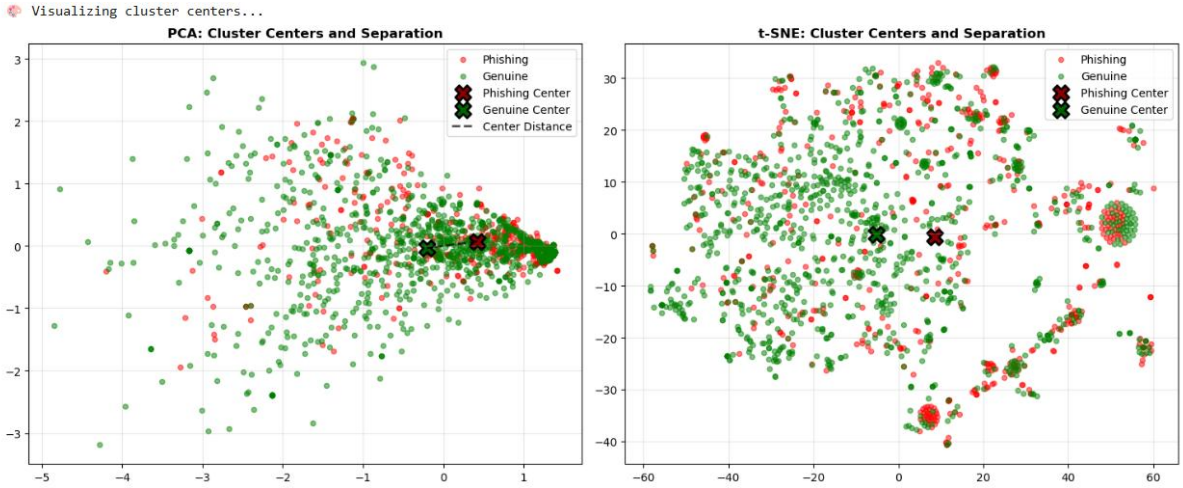

Figure 4:Appendix d